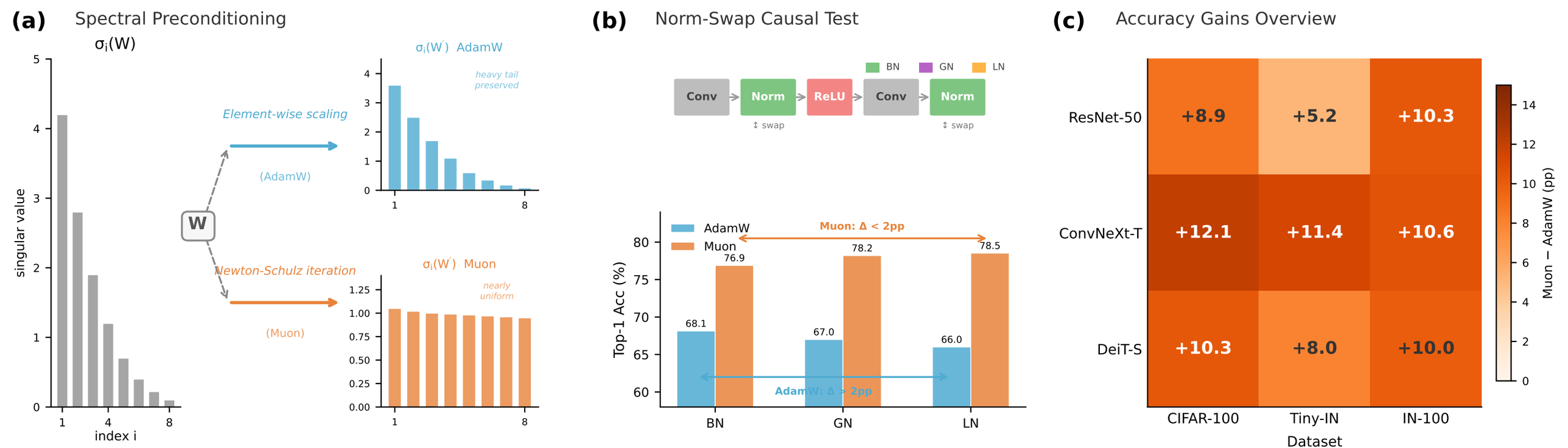
(a)
Spectral Preconditioning
σi(W)
singular value
index i
W
Element-wise scaling
(AdamW)
Newton-Schulz iteration
(Muon)
σi(W′) AdamW
heavy tail preserved
σi(W′) Muon
nearly uniform
(b)
Norm-Swap Causal Test
Conv
Norm
ReLU
Conv
Norm
BN
GN
LN
↕ swap
↕ swap
AdamW
Muon
Top-1 Acc (%)
Muon: Δ < 2pp
AdamW: Δ > 2pp
76.9
68.1
78.2
67.0
78.5
66.0
BN
GN
LN
(c)
Accuracy Gains Overview
ResNet-50
ConvNeXt-T
DeiT-S
+8.9
+5.2
+10.3
+12.1
+11.4
+10.6
+10.3
+8.0
+10.0
CIFAR-100
Tiny-IN
IN-100
Dataset
Muon − AdamW (pp)